

5. METHODS OF PROOF

Logical arguments are used to give proofs of theorems. In computing such proofs are essential in the design and verification of algorithms. The most common types of proofs are the ones where we try to establish the truth of a proposition of the form $(P \Rightarrow Q)$. There are several methods of proof. We are going to examine three of them, with more emphasis given to Mathematical Induction.

Direct Argument:

To prove a proposition of the form $(P \Rightarrow Q)$ by direct argument, we assume P is true, and through a series of mathematically correct steps we show that Q is true.

The examples we are going to see are introductory and most of the times we will be able to use more than one method. We will continuously be referring to odd and even integers all the time, so the following will be useful:

- If a is even we write $a = 2k$ for some integer k .
- If a is odd we write $a = 2k + 1$ for some integer k .
- If a is divisible by a number b we write $a = kb$ for some integer k .

Example 1:

Use a direct method of proof to show that if x and y are odd integers, then xy is also odd.

Solution:

Since x and y are both odd we write $x = 2m + 1$ and $y = 2n + 1$ for some integers m and n . Then, $xy = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1 = 2k + 1$, where $k = 2mn + m + n$. Hence, xy is also odd.

Example 2:

Use a direct method of proof to show that if x and y are even integers, then $3x^2 + 5xy - 6y$ is a multiple of 4.

Solution:

Since x and y are both even we write $x = 2m$ and $y = 2n$ for some integers m and n . Then,

$$\begin{aligned} 3x^2 + 5xy - 6y &= 3(2m)^2 + 5(2m)(2n) - 6(2n) \\ &= 3(4m^2) + 20mn - 12n \\ &= 12m^2 + 20mn - 12n \\ &= 4(3m^2 + 5mn - 3n) \\ &= 4k \end{aligned}$$

where $k = 3m^2 + 5mn - 3n$. Hence, $3x^2 + 5xy - 6y$ is a multiple of 4.

Proof by Contradiction:

To prove a proposition by contradiction, we assume that the statement of the proposition is false, and through mathematically sound steps derive a contradiction (show that something is wrong with our assumption). Hence, we conclude that the statement is true.

Example 3:

Prove by contradiction that there does not exist a largest number.

Solution:

Assume for the purpose of contradiction, there is a largest number, and suppose that this largest number is A . But then $A + 1 > A$ which leads to a contradiction. Hence, we conclude that there does not exist a largest number.

Example 4:

Prove by contradiction that if $x^2 = 2$ then x is not a fraction. (In other words $\sqrt{2}$ is not a rational number)

Solution: Assume for the purpose of contradiction, that x is a fraction and write $x = \frac{m}{n}$, where m and n are integers, $n \neq 0$ and m and n have no common factors (if they do have common factors then cancel them out). Since $x^2 = 2$ we have $(m/n)^2 = 2$, therefore, $m^2 = 2n^2$. This means that m^2 is even, which in turn means that m is even as well. Hence we write $m = 2p$. Substituting back into our equation we get $(2p)^2 = 2n^2 \Rightarrow 4p^2 = 2n^2 \Rightarrow 2p^2 = n^2$. But this means that n^2 and in turn n is an even number as well. This contradicts our initial assertion that m and n had no common factors, since they are both divisible by 2. Hence, our initial assumption that if $x^2 = 2$ then x is a fraction must be wrong. Hence, we conclude that if $x^2 = 2$ then x is not a fraction.

The Principle of Mathematical Induction:

Suppose that $P(n)$ is a statement involving a general positive integer n . Then $P(n)$ is true for all positive integers if:

1. $P(1)$ is true (we call this the **base case**) and
2. $P(k) \Rightarrow P(k+1)$ for all positive integers k (we call this the **inductive step**)

The base case shows that the proposition is true for $n = 1$. The inductive step shows that if the proposition is true for one integer $n = k$, then it is true for its successive integer as well $n = k + 1$. The two steps together show that the proposition is true for all integers n .

Example 5:

Prove by induction that

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

for all $n \geq 1$.

Notation: $\sum_{i=1}^n i = 1 + 2 + 3 + \dots + n$

Solution:

Note that $P(n)$ is the statement

$$\sum_{i=1}^n i = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Then $P(1)$ is the statement:

$$\sum_{i=1}^1 i = 1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1$$

which is true. For the inductive step, we assume that

$$\sum_{i=1}^k i = 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

and we want to show that

$$\sum_{i=1}^{k+1} i = 1 + 2 + 3 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2}$$

Therefore:

$$\begin{aligned}
 1 + 2 + 3 + \dots + k + (k + 1) &= (1 + 2 + 3 + \dots + k) + (k + 1) \\
 &= \frac{k(k + 1)}{2} + (k + 1) && \text{because of the inductive step} \\
 &= \frac{k(k + 1) + 2(k + 1)}{2} \\
 &= \frac{k^2 + k + 2k + 2}{2} \\
 &= \frac{k^2 + 3k + 2}{2} \\
 &= \frac{(k + 1)(k + 2)}{2}
 \end{aligned}$$

Hence, $P(k) \Rightarrow P(k + 1)$ for all positive integers k . Therefore, by induction, we conclude that

$$\sum_{i=1}^n i = \frac{n(n + 1)}{2}$$

is true for all positive integers $n \geq 1$. □

Example 6:

Prove by induction that

$$\sum_{i=1}^n i(i+1)(i+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

for all $n \geq 1$.

Solution: $P(n)$ is the statement:

$$\sum_{i=1}^n i(i+1)(i+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

Then $P(1)$ is the statement:

$$6 = 1(1+1)(1+2) = \frac{1(1+1)(1+2)(1+3)}{4} = \frac{1 \cdot 2 \cdot 3 \cdot 4}{4} = 6$$

Hence, $P(1)$ is true. For our inductive step, assume that

$$\sum_{i=1}^k i(i+1)(i+2) = \frac{k(k+1)(k+2)(k+3)}{4}$$

and let us show that $P(k+1)$ is true as well. Note that $P(k+1)$ is the following statement.

$$\sum_{i=1}^{k+1} i(i+1)(i+2) = \frac{(k+1)(k+2)(k+3)(k+4)}{4}$$

Hence,

$$\begin{aligned} \sum_{i=1}^{k+1} i(i+1)(i+2) &= \left(\sum_{i=1}^k i(i+1)(i+2) \right) + (k+1)(k+2)(k+3) \\ &= \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3) && \text{inductive step} \\ &= \frac{k(k+1)(k+2)(k+3) + 4(k+1)(k+2)(k+3)}{4} \\ &= \frac{[(k+1)(k+2)(k+3)](k+4)}{4} \\ &= \frac{(k+1)(k+2)(k+3)(k+4)}{4} \end{aligned}$$

We showed that $P(k) \Rightarrow P(k+1)$ for all integers $k \geq 1$, therefore, by induction

$$\sum_{i=1}^n i(i+1)(i+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

for all $n \geq 1$. □

Example 7:

Prove by induction that

$$\sum_{i=1}^n 3^i = \frac{3^{n+1} - 3}{2}$$

for all $n \geq 1$.

Solution:

$P(n)$ is the statement

$$\sum_{i=1}^n 3^i = 3^1 + 3^2 + 3^3 + \dots + 3^n = \frac{3^{n+1} - 3}{2}$$

Then $P(1)$ is the statement:

$$\sum_{i=1}^1 3^i = 3^1 = 3 = \frac{3^{1+1} - 3}{2} = \frac{9 - 3}{2} = \frac{6}{2} = 3$$

which is true. For the inductive step, we assume that

$$\sum_{i=1}^k 3^i = 3^1 + 3^2 + 3^3 + \dots + 3^k = \frac{3^{k+1} - 3}{2}$$

and we want to show that

$$\sum_{i=1}^{k+1} 3^i = 3^1 + 3^2 + 3^3 + \dots + 3^k + 3^{k+1} = \frac{3^{k+2} - 3}{2}$$

Therefore:

$$\begin{aligned} 3^1 + 3^2 + 3^3 + \dots + 3^k + 3^{k+1} &= (3^1 + 3^2 + 3^3 + \dots + 3^k) + 3^{k+1} \\ &= \frac{3^{k+1} - 3}{2} + 3^{k+1} && \text{because of the inductive step} \\ &= \frac{3^{k+1} - 3 + 2 \cdot 3^{k+1}}{2} \\ &= \frac{3 \cdot 3^{k+1} - 3}{2} \\ &= \frac{3^{k+2} - 3}{2} \end{aligned}$$

Hence, $P(k) \Rightarrow P(k+1)$ for all positive integers k . Therefore, by induction, we conclude that

$$\sum_{i=1}^n 3^i = 3^1 + 3^2 + 3^3 + \dots + 3^n = \frac{3^{n+1} - 3}{2}$$

is true for all positive integers $n \geq 1$. □

Example 8: Prove, by induction, that $(n^2 + n)$ is even for all $n \geq 1$.

Solution: Let us formulate $P(n)$ first. $P(n)$ is the statement: “ $(n^2 + n)$ is even”. In other words $(n^2 + n) = 2b$ for some integer b .

The base case $P(1)$ is the statement $(1^2 + 1)$ is even. Since $1^2 + 1 = 1 + 1 = 2$, we see that $P(1)$ is true.

Now for the inductive step, assume that $(k^2 + k) = 2b$ (for some integer b) and let us show that $((k + 1)^2 + (k + 1))$ is also even

$$\begin{aligned} ((k + 1)^2 + (k + 1)) &= (k^2 + 2k + 1) + (k + 1) \\ &= (k^2 + k) + (2k + 2) \\ &= 2b + 2(k + 1) && \text{because of the inductive step} \\ &= 2(b + k + 1) \\ &= 2m \end{aligned}$$

We showed that if $(k^2 + k)$ is even then so is $((k + 1)^2 + (k + 1))$ for any integer k . Therefore, by induction, $(n^2 + n)$ is even for all $n \geq 1$. $\square$

Example 9:

Prove, by induction, that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Solution:

Let us formulate $P(n)$ first. $P(n)$ is the statement: “ $7^n - 1$ is divisible by 6”. In other words $7^n - 1 = 6b$ for some integer b .

The base case $P(1)$ is the statement $7^1 - 1$ is divisible by 6. Since $7^1 - 1 = 7 - 1 = 6$, we see that $P(1)$ is true.

Now for the inductive step, assume that $7^k - 1 = 6b$ and let us show that $7^{k+1} - 1$ is also divisible by 6

$$\begin{aligned} 7^{k+1} - 1 &= 7 \cdot 7^k - 1 \\ &= 7(7^k - 1) + 7 - 1 \\ &= 7(6b) + 6 && \text{because of the inductive step} \\ &= 6(7b + 1) \end{aligned}$$

We showed that if $7^k - 1$ is divisible by 6 then so is $7^{k+1} - 1$ for any integer k . Therefore, by induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$. $\square$